Fairness in AI

Lecture 2: Biases

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Today's lecture

- Motivation
- Human biases
- Data-related biases
- Task-oriented bias analysis
- Evaluation biases

- Course link:

https://docs.google.com/presentation/d/17Z0TbFe4NDfSt4PPY7ERSOxsqcTRnLA7EWvhl_rwGRw/edit?slide=id.p#slide=id.p

Motivation

“Bias, man. It’s everywhere”

Arthur Mello • 3rd+ **+ Follow** ···
Data scientist and educator
2d • Edited •

🔥 How to Become an AI Expert in 5 minutes

1. Say “garbage in, garbage out.” It means nothing, but it sounds cool.

2. Ask what LLM they’re using. It’s absolutely irrelevant, but it makes you sound like you know what you’re talking about. Bonus points if you tilt your head and say, “Oh, GPT-4? Interesting choice.”

3. Talk about “the black box problem.” You don’t need to know what it actually means—just sigh heavily and say, “That’s the thing with AI, isn’t it? It’s all a black box.” Everyone will nod solemnly.

4. Use “fine-tuning” in every other sentence. You don’t need to understand the difference between fine-tuning and prompt engineering. Just say things like, “Well, it’s not performing well because they probably didn’t fine-tune it properly.”

5. Mention ethical concerns vaguely. Say, “We need to consider ethical issues here too.” But never specify which ethical problems you’re referring to. If pressed, just shake your head and mutter: “Bias, man. It’s everywhere.”

6. Throw in words like “unstructured data” and “scalability.” For example: “The issue is that most companies can’t handle unstructured data at scale.” People will assume you’re deep into AI architecture, even if you’re just deep into Twitter.

7. Make wild predictions about AI taking over. Alternate between saying, “AI will replace 80% of jobs in the next five years,” and, “AI is overhyped; it’ll never truly replace humans.” Contradict yourself freely—it’s all part of the mystique.

8. End every conversation with “We’re still in the early days of AI.” It’s the perfect way to sound both optimistic and profound while giving yourself an exit strategy.

Congratulations! You are now ready to dazzle coworkers and intimidate LinkedIn connections with your newfound AI “expertise.”

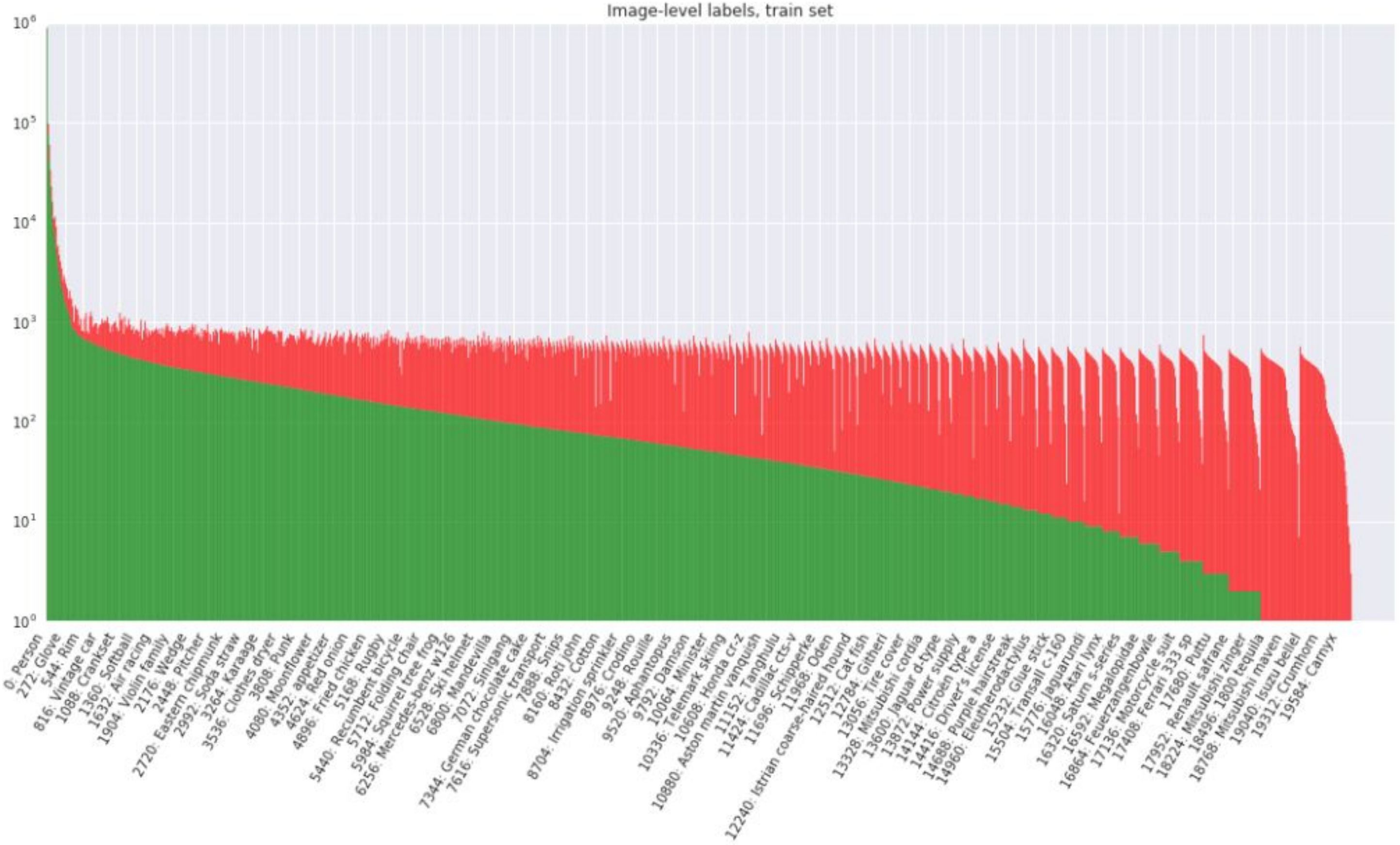
Just remember: when in doubt, say “It’s all about the data.” No one will argue.

What is important in the following work review?

You are able to finish your assignments in a timely manner and this is appreciated. Most of the code you write works as expected and is well documented. However, the class structure is not always clear and some of the methods are not completely optimized.

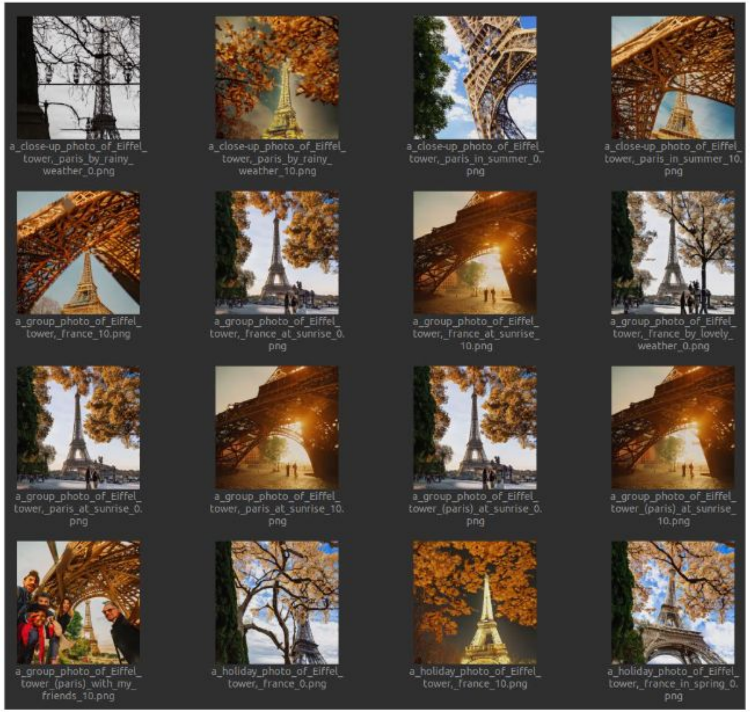
OpenImages v3 train set distribution

Log scale!

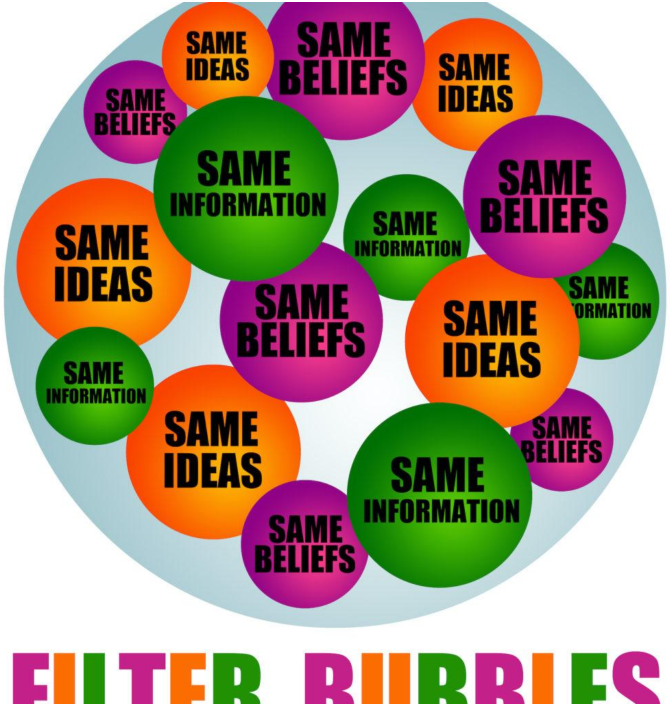


Task-oriented bias analysis

“Prototypical” viewpoints and backgrounds in image generation

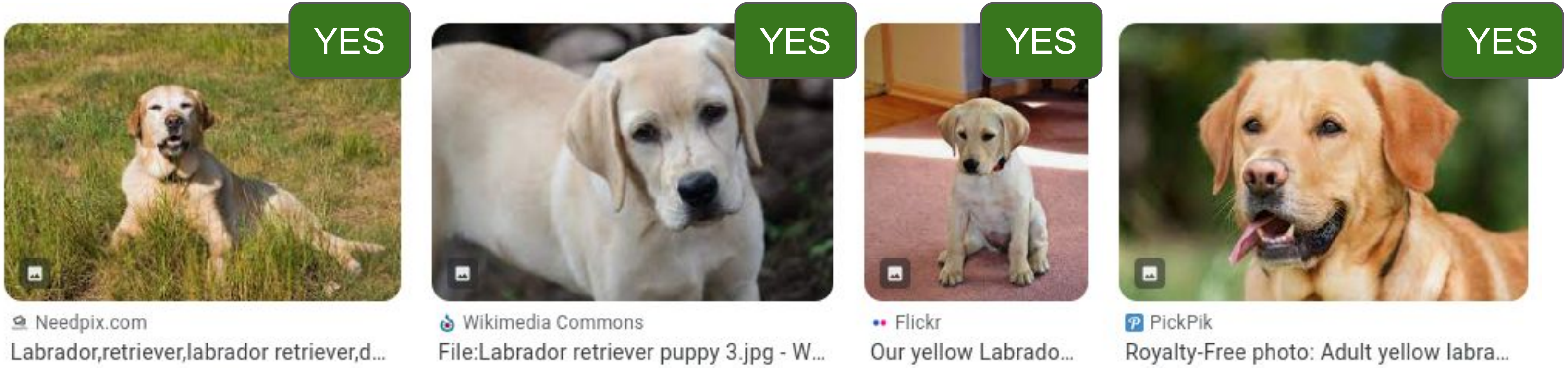


Filter bubbles created by personalized recommendations?



Cherry picking in evaluation

- Is this a dog?
- Test images



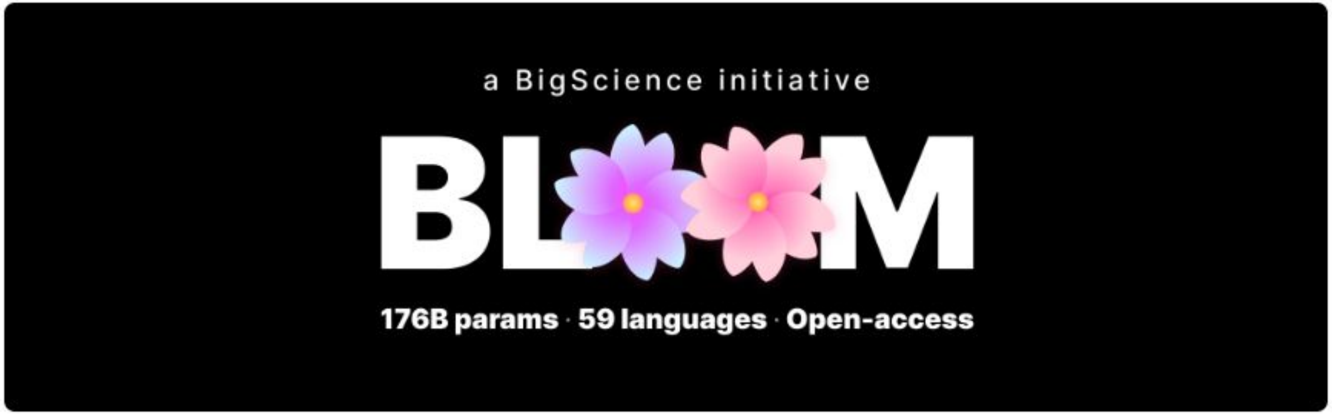
- Perfect classifier

Cherry picking in evaluation

- Is this a dog?
- Complete test set

- ~~Perfect classifier~~

Infrastructure matters



Here's a quick summary of project:

Hardware	384 80GB A100 GPUs
Software	Megatron-DeepSpeed
Architecture	GPT3 w/ extras
Dataset	350B tokens of 59 Languages
Training time	3.5 months

Biases in AI

- Biases can occur all steps of AI pipelines
- Potentially catastrophic effects when they are combined
- They are the main cause of unfairness
- Some of them are obvious, others are more implicit
- What can be done?
    - Be aware of the existence of multiple biases (today)
    - Understand and counter biases (next time)

Human biases

Gender

Gender bias in the European AI workforce

Figure 1. Gender gap among AI professionals in the EU (%)

Gender bias factors

- Lack of diversity in training data and workforce
- Bias in society - gender stereotyping
- Bias in ~~algorithm~~ due to data -> model
  - Developer biases (conscious or not)
  - Gender bias in everyday language
  - Language discrimination in gender-related data
- Economic factors
- Biased behaviors and decisions
  - Human biases transferred in ~~algorithms~~ -> models

Top “voices” in AI (last year)

Here’s a concise list of the top 10 wealthiest AI entrepreneurs:

1. **Elon Musk** ($172B) – Tesla, Neuralink; AI in autonomous driving and brain tech.
2. **Jeff Bezos** ($178B) – Amazon; AI in AWS and Alexa.
3. **Larry Page** ($125B) – Google; AI via DeepMind and Waymo.
4. **Sergey Brin** ($105B) – Google; AI-driven projects at Alphabet.
5. **Mark Zuckerberg** ($134B) – Meta; AI across Facebook and VR.
6. **Jensen Huang** ($48.6B) – NVIDIA; AI hardware leader.
7. **Satya Nadella** ($320M) – Microsoft; AI in Azure and acquisitions.
8. **Tim Cook** ($1.5B) – Apple; AI in Siri and devices.
9. **Sundar Pichai** ($600M) – Google; advancing AI across products.
10. **Reid Hoffman** ($2.5B) – LinkedIn; AI investments and innovations.

Google Scholar | label:artificial_intelligence

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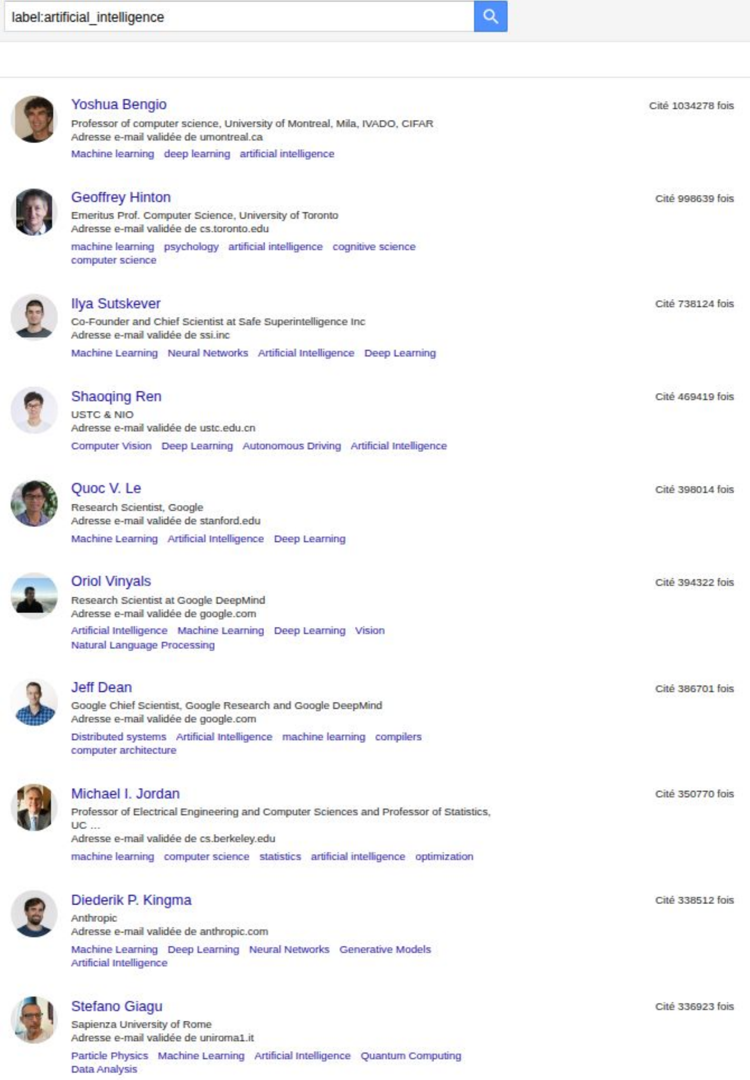
Top “voices” in AI (now)

Ranked list: Wealthiest tech entrepreneurs with significant AI involvement

(Net worth on the same line; estimates rounded, early-2026 levels; AI is a material part of each business)

1. **Elon Musk (USA)** — ~$650–700B
   Tesla (autonomy), xAI, AI-driven robotics and optimization
2. **Larry Ellison (USA)** — ~$240–250B
   Oracle Cloud Infrastructure, enterprise AI platforms
3. **Larry Page (USA)** — ~$240–260B
   Alphabet / Google, DeepMind, Gemini, AI-powered search & cloud
4. **Sergey Brin (USA)** — ~$230–250B
   Alphabet / Google, AI research and deployment at scale
5. **Jeff Bezos (USA)** — ~$220–250B
   Amazon, AWS AI services, logistics and automation
6. **Mark Zuckerberg (USA)** — ~$210–230B
   Meta AI, LLaMA models, recommendation and generative systems
7. **Zhang Yiming (China)** — ~$180–200B
   ByteDance (TikTok/Douyin), world-leading AI recommendation engines
8. **Jensen Huang (Taiwan/USA)** — ~$150–170B
   NVIDIA, dominant AI compute and accelerator ecosystem
9. **Pony Ma (Ma Huateng) (China)** — ~$45–55B
   Tencent, AI in gaming, payments, social platforms, foundation models
10. **Jack Ma (China)** — ~$35–45B
   Alibaba, AI-driven commerce, cloud AI, logistics optimization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Lost in translation biases

Google Translate

DeepL

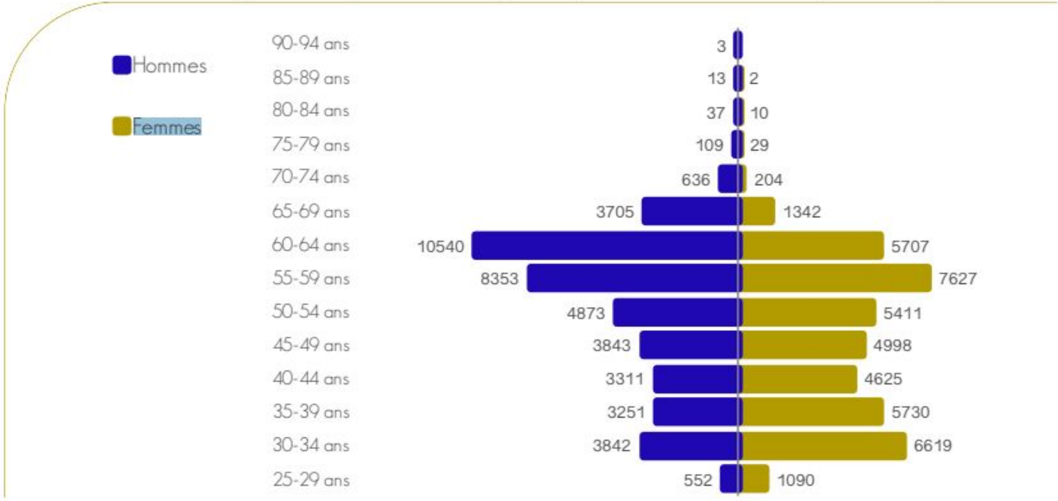
Some facts

	Female	Male
Global	46	54
<60 years	52	48
Prof.	**14**	**86**
Ass. Prof.	**47**	**53**

Doctors' gender distribution in France

General practitioners' gender distribution in France

Figure 13. Pyramides des âges des généralistes en activité régulire au 1er janvier 2020



French COVID19 experts following ChatGPT (2024)

1. Dr. Didier Raoult - A microbiologist and infectious disease specialist known for his research on hydroxychloroquine and azithromycin as potential COVID-19 treatments.
2. Dr. Jean-François Delfraissy - An immunologist and director of the Scientific Council, which advised the French government on COVID-19 measures.
3. Dr. Karine Lacombe - An infectious disease specialist and epidemiologist who has been actively involved in research and public health efforts related to COVID-19.
4. Dr. Arnaud Fontanet - An epidemiologist and professor at the Pasteur Institute in Paris, known for his expertise in the spread and control of infectious diseases.
5. Dr. Anne-Claude Crémieux - An infectious disease specialist and professor who has provided insights into COVID-19 treatment and management.
6. Dr. Yazdan Yazdanpanah - An infectious disease specialist and epidemiologist known for his research on HIV and involvement in COVID-19 response efforts.
7. Dr. Elisabeth Bouvet - A virologist and professor at the University of Paris-Saclay who has contributed to COVID-19 research.
8. Dr. Bruno Lina - A virologist and expert in respiratory viruses who has provided valuable insights into COVID-19.
9. Dr. Bruno Hoen - An infectious disease specialist and epidemiologist with expertise in tropical medicine, he has contributed to COVID-19 research and public health efforts.
10. Dr. François Balloux - Although not French, Dr. Balloux is a prominent geneticist and epidemiologist based at the University College London, and his work has been influential in understanding the genetic aspects of COVID-19 transmission.

Addressing gender bias

- Fair and ethical design and implementation of AI tech
  - Policies, fair testing, transparency
- Reducing bias from the ~~algorithm~~ -> design, data, model
- Ensuring diversity in AI development
  - Increasing the number of female AI developers
  - Interdisciplinary teams including tech & social science experts
  - Neutralization of language in data and models

Addressing gender bias

- Misuse of "algorithm" here
- It's often a data-related problem

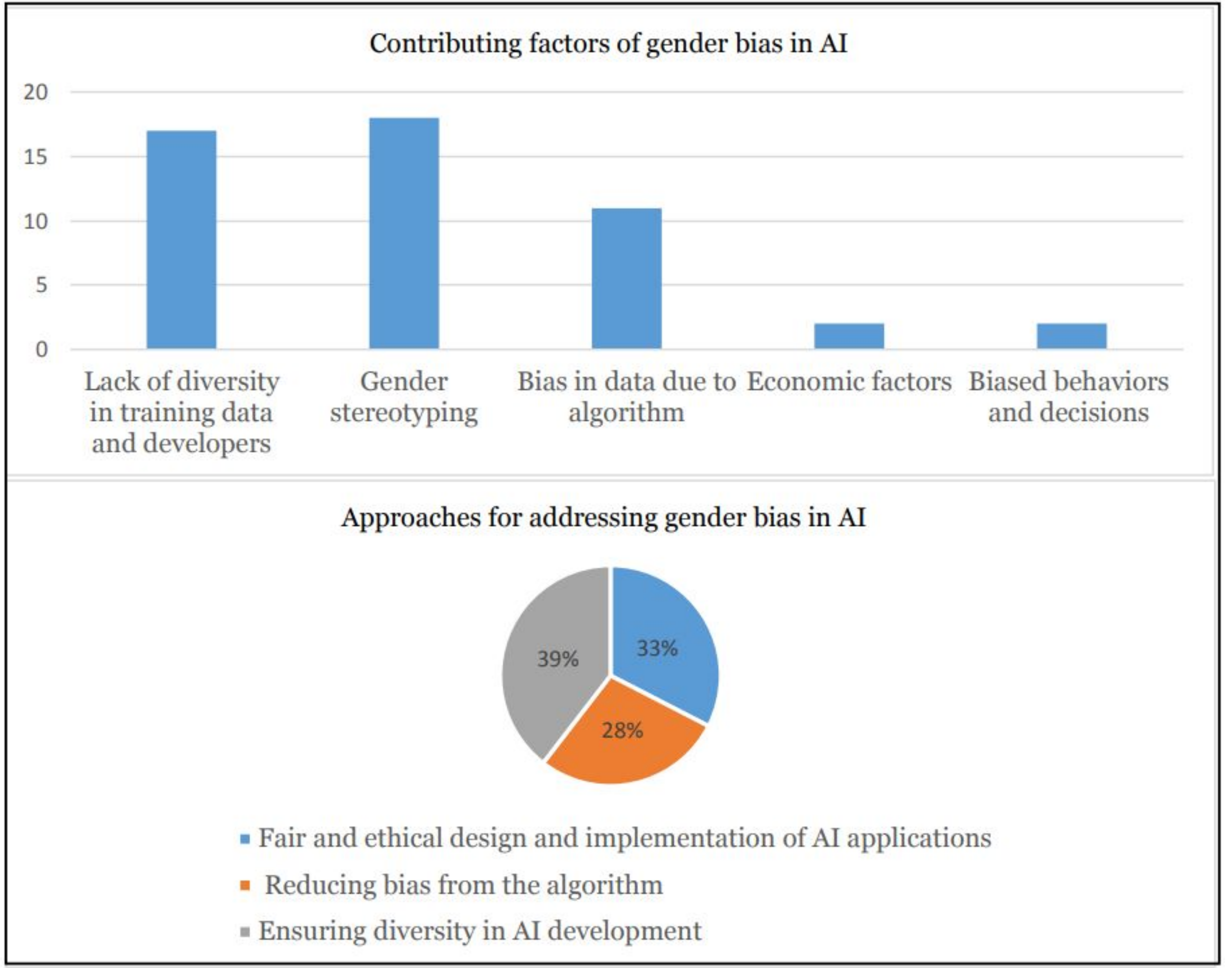


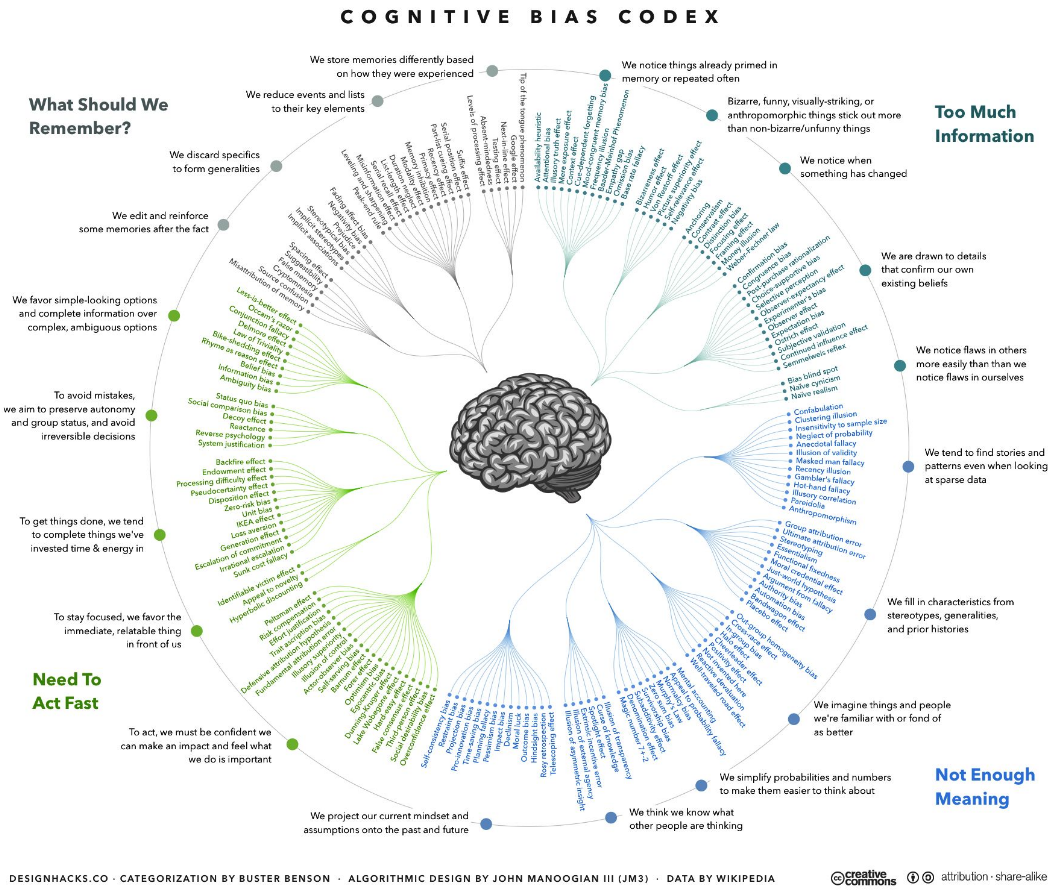
Figure 2. Graphical representation of the themes of contributing factors and approaches for addressing gender bias in AI.

Human biases

Cognitive biases

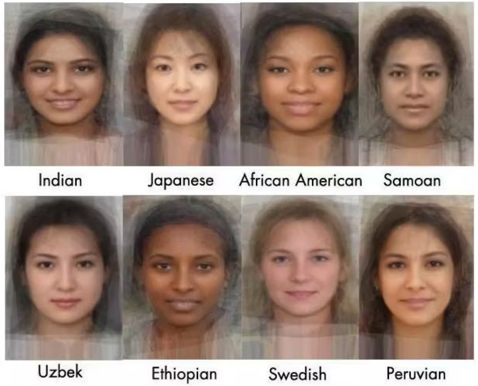
Cognitive biases

- Crowded space
- Due to heuristics applied by our brains
- Effects on our everyday lives, including professional activities

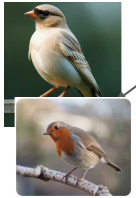


Prototype theory

- Remember the animals and vehicles?
- Prototypes - average representations of categories
  - Efficient representation of the world
  - Practical example: cluster centroids
- Membership
  - Category members are not born equal
  - Nor are they equally frequent
- Variability



Avg woman faces



bird

Cultural: personal/social experience

Temporal evolution

1896 telephone from Sweden

An old rotary dial telephone

AT&T push button telephone made by Western Electric, model 2500 DMG black, 1980

Effects of cognitive biases on the interpretation of rule-based ML methods

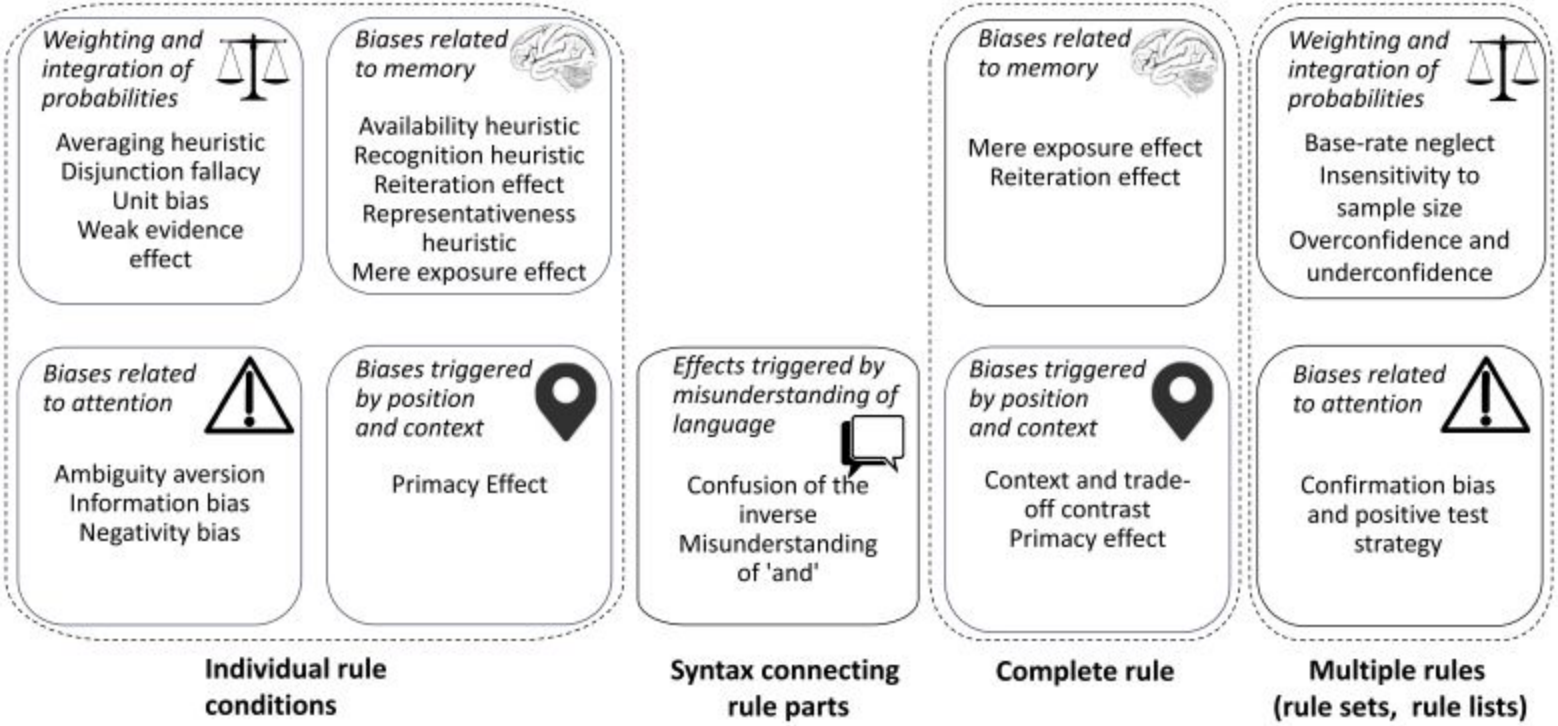


Fig. 4. Cognitive biases grouped by the affected or triggering rule model element and by their underlying mechanism.

Effects of cognitive biases on the interpretation of rule-based ML methods

Phenomenon	Implications for rule-learning	Debiasing technique
Conjunction Fallacy and Representativeness Heuristic **(Section 5.1): Using prototypicality for judgment of probability, which may lead people to judge a conjunction of two statements to be more probable than one of the two statements**	Overestimate the probability of condition representative of rule consequent	Use natural frequencies instead of ratios or probabilities, engage System 2, provide training in logic
Insensitivity to Sample Size **(Section 5.6): The tendency to underestimate the benefits of larger samples**	Analyst does not realize the increased reliability of confidence estimate with increasing value of support	Present support as absolute number; show confidence (reliability) intervals for the value of confidence

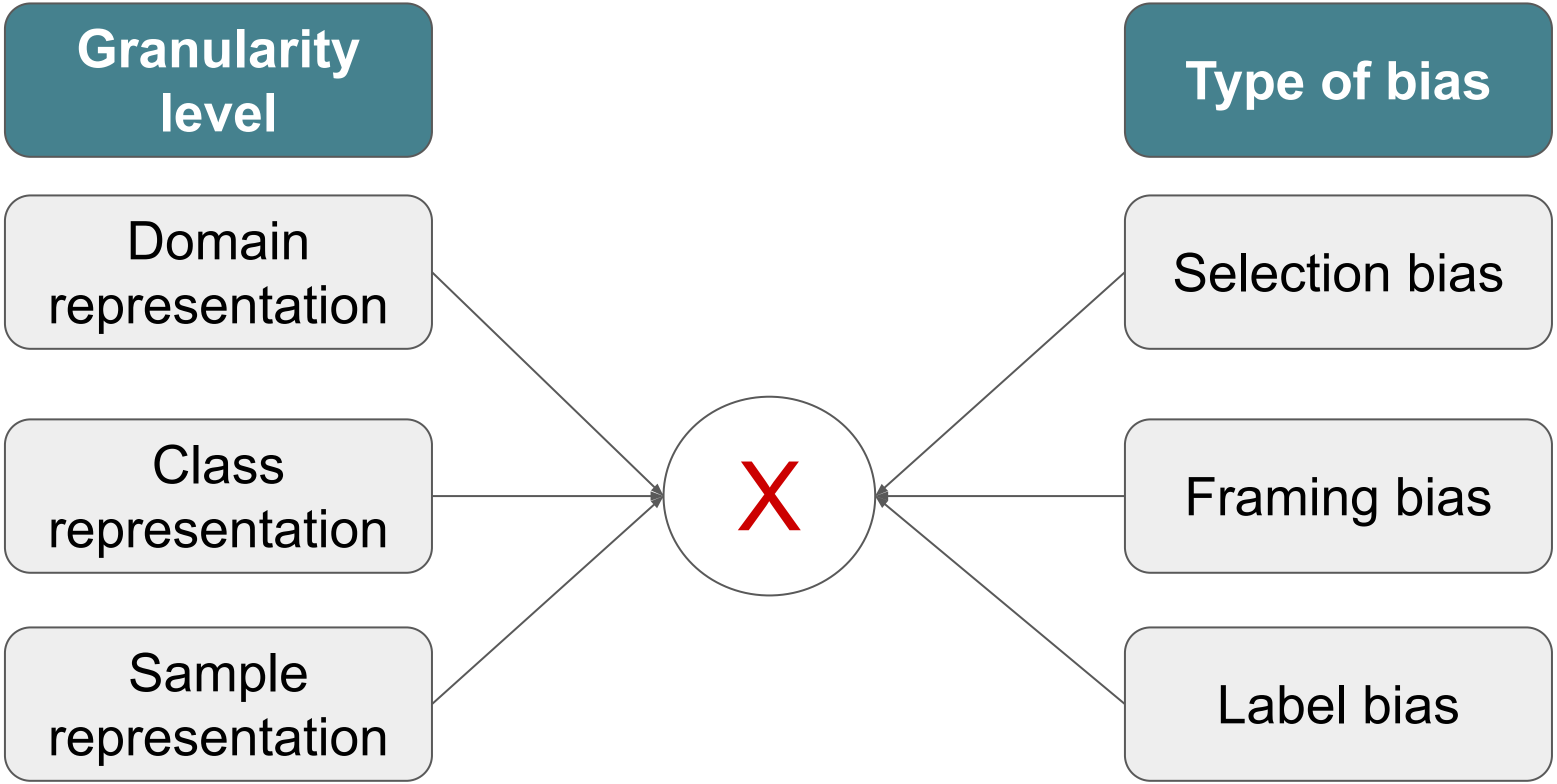
Effects of cognitive biases on the interpretation of rule-based ML methods

Phenomenon	Implications for rule-learning	Debiasing technique
Availability heuristic **(Section 5.8): Perceived frequency of a class is determined by the ease with which its instances come to mind**	Rules for which instances are easily recalled are considered as more plausible	Explain why for some combinations of conditions the instances are easily recalled and not for others
Information Bias **(Section 5.13): The tendency to seek information even when it is not relevant or helpful**	Belief that more information (rules, conditions) will improve decision making even if it is irrelevant	Communicate attribute importance
Negativity Bias **(Section 5.17): The tendency to overweight negative information over positive information of the same strength**	Words with negative valence in the rule make it appear more important	Review words with negative valence in data, and possibly replace with neutral alternatives

Data-related biases

Bias types

Bias amplification effects



Domain representation

Text -
Wikipedia by occupation
(~2010)

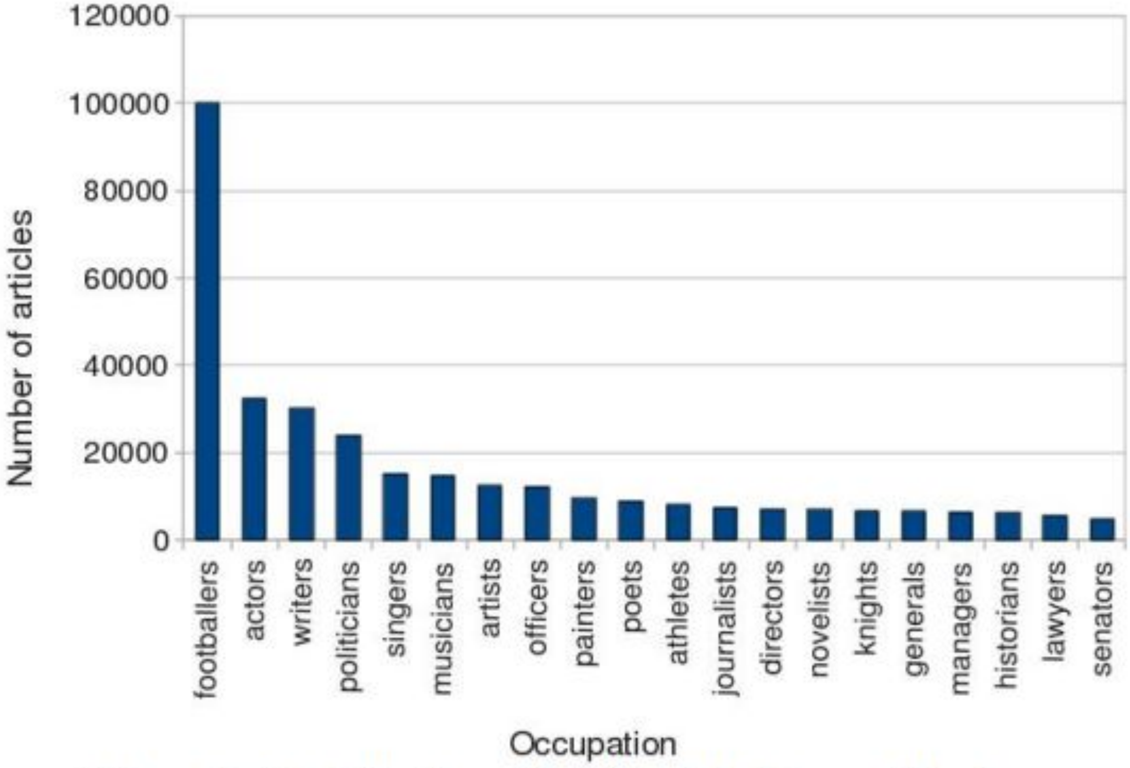


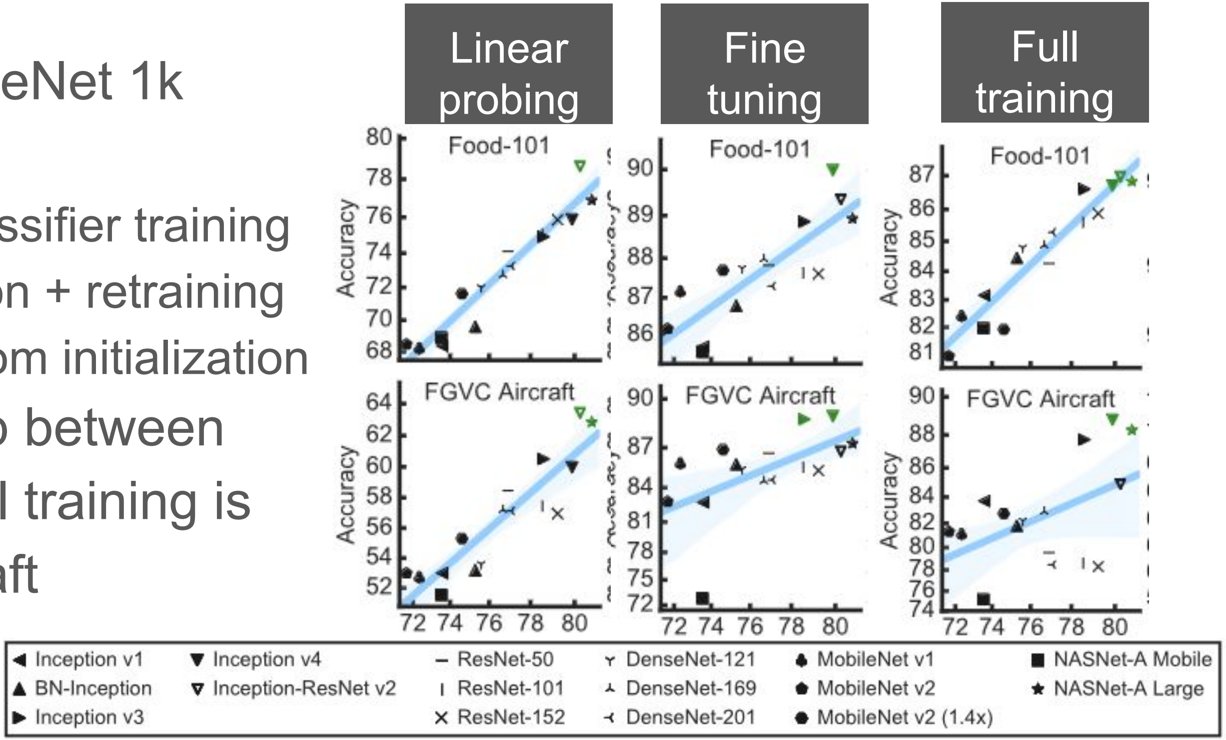
Figure 5. Distribution of Wikipedia biographies by occupation.

Domain representation

Images - visually coherent Flickr groups (~2014)

Effect of domain representation in training sets

- Pretraining with ImageNet 1k
- Training strategies
  - Probing - linear classifier training
  - Tuning - initialization + retraining
  - Full training - random initialization
- The performance gap between linear probing and full training is much larger for Aircraft



Biases in visual datasets

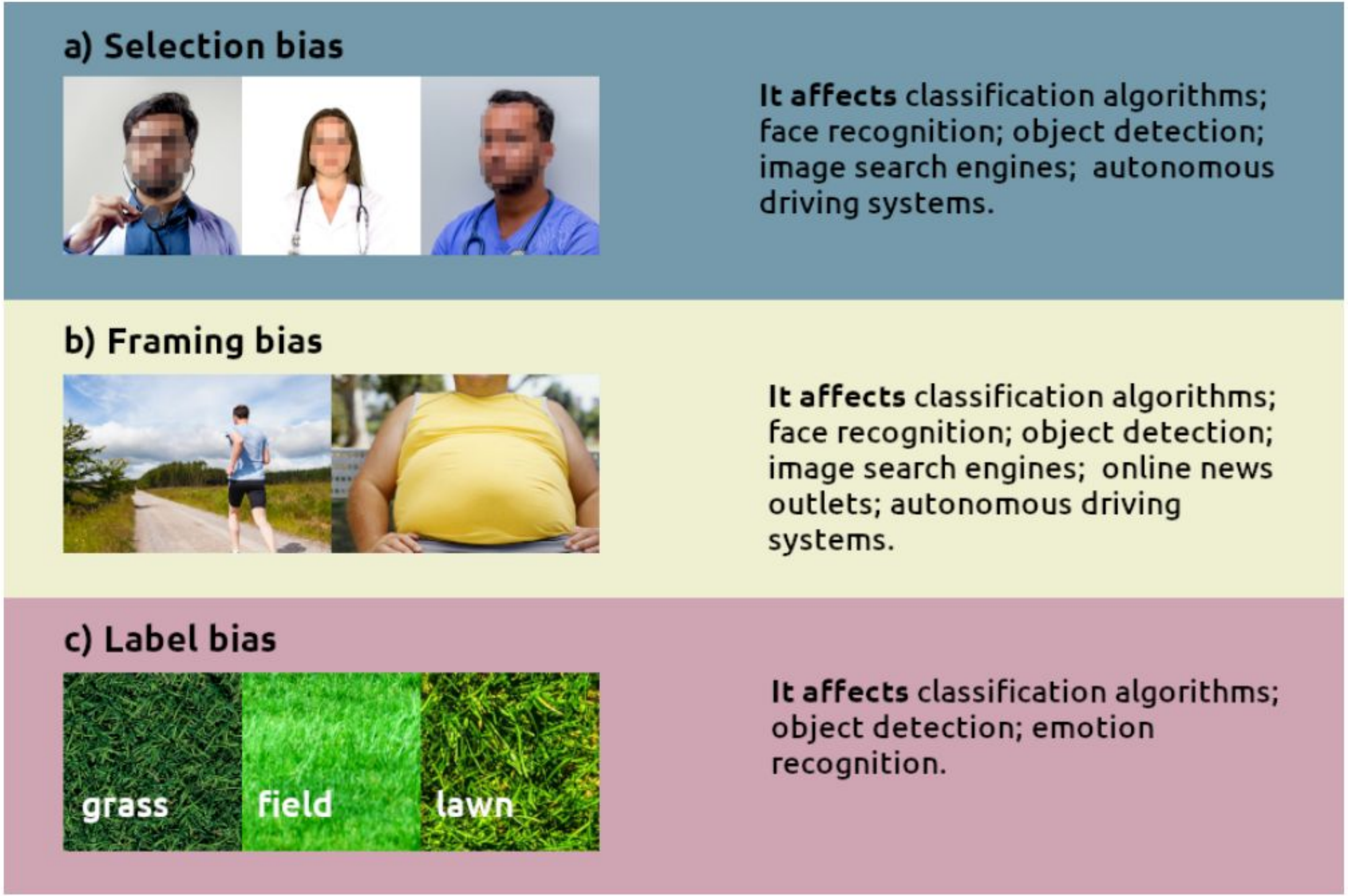
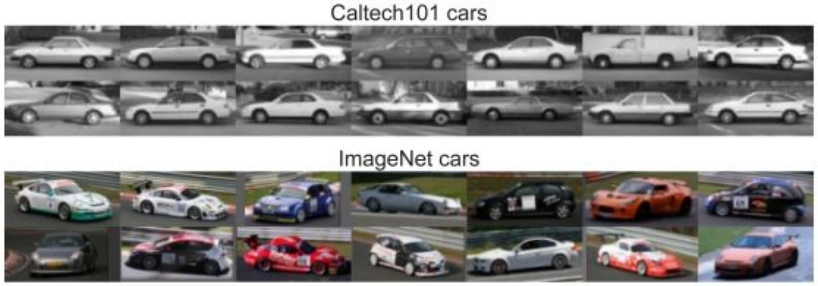


Figure 1: Examples of selection, framing and label bias. On the right, a list of applications that can be affected by each type of bias.

Selection bias

> *We call* selection bias *any disparities or associations created as a result of the process by which subjects are included in a visual dataset.*

- Training and test data mismatch
- Rooted in a sampling problem
  - Who took the photos?
  - What text is associated with them?
  - How were queries formulated during data collection?
- Main negative effects in applications that involve people
  - Autonomous driving
  - Face recognition
- Data balancing is needed but sometimes insufficient!



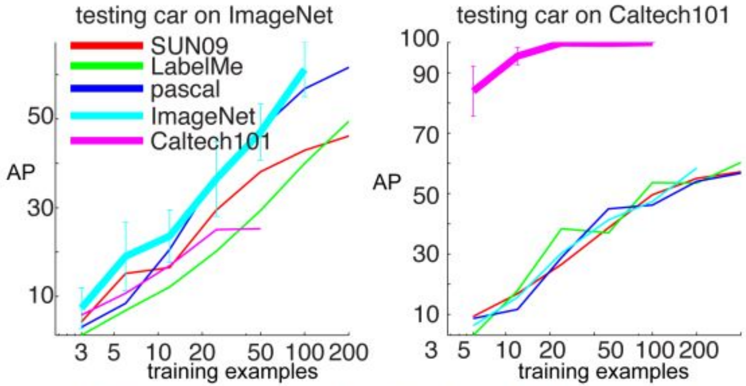
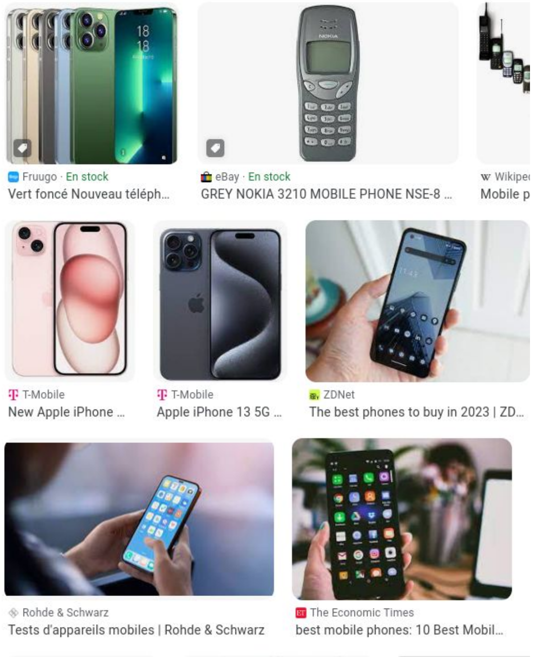


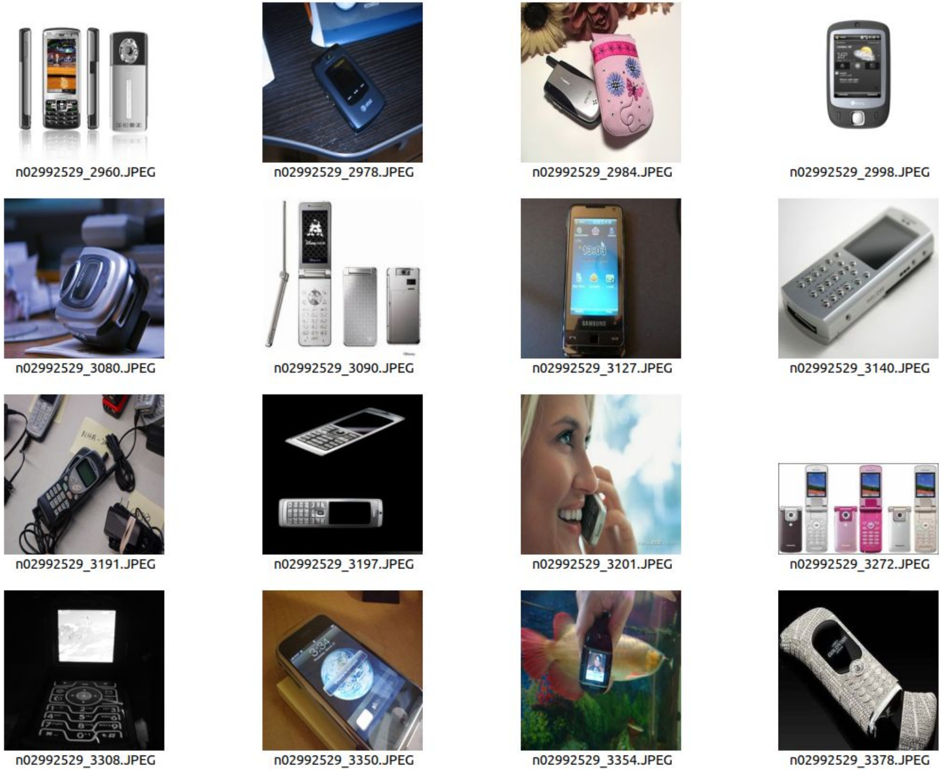
Figure 6. Cross-dataset generalization for "car" detection as function of training data

Temporal selection bias

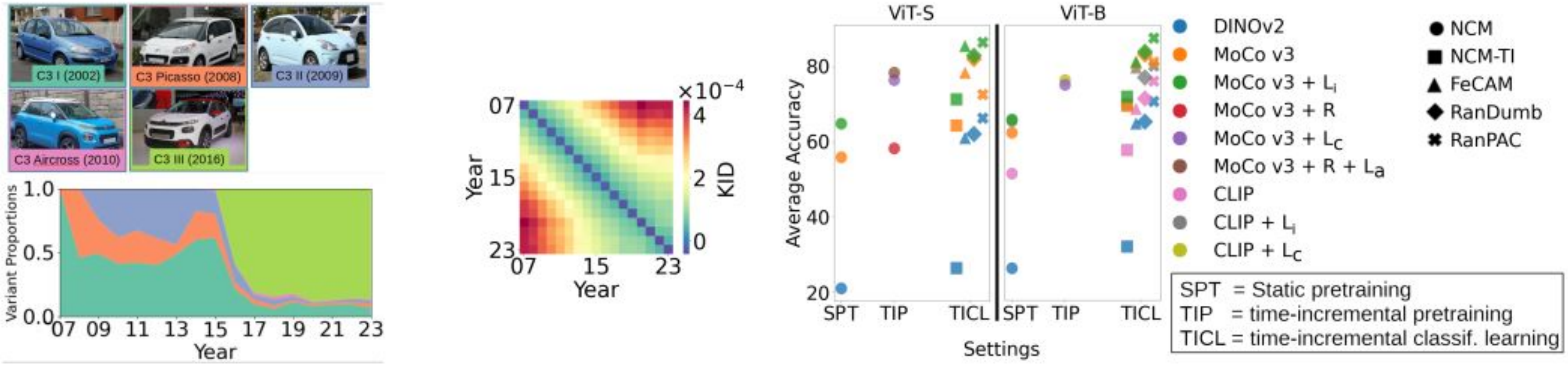
Test - “Mobile phone” in Google Images (2024)



Train - “mobile phone” aka “cellular phone” in ImageNet (<= 2008)



CaMiT: time-aware car model dataset



(a) (b) (c)

Figure 1: Illustration of CaMiT content and experiments. (a) exemplifies the *Citroën C3* data shift in time. (b) shows that the averaged kernel-inception distance between class embeddings increases with time. (c) highlights the positive effects of time-aware classification over static pretraining.

Cultural selection bias - “toy”

?

?

?

Cultural selection bias - “toy”

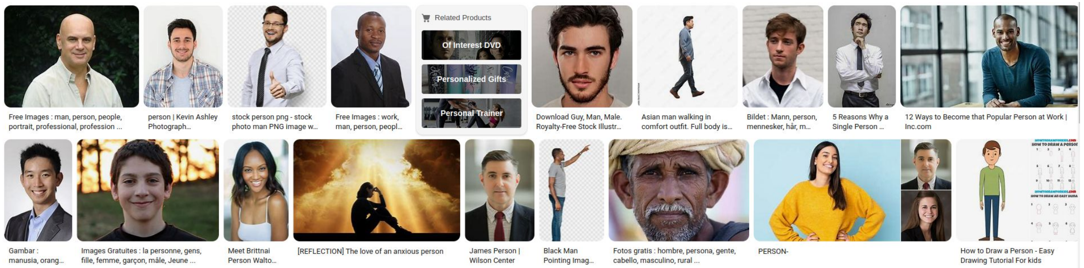
Japan

Russia

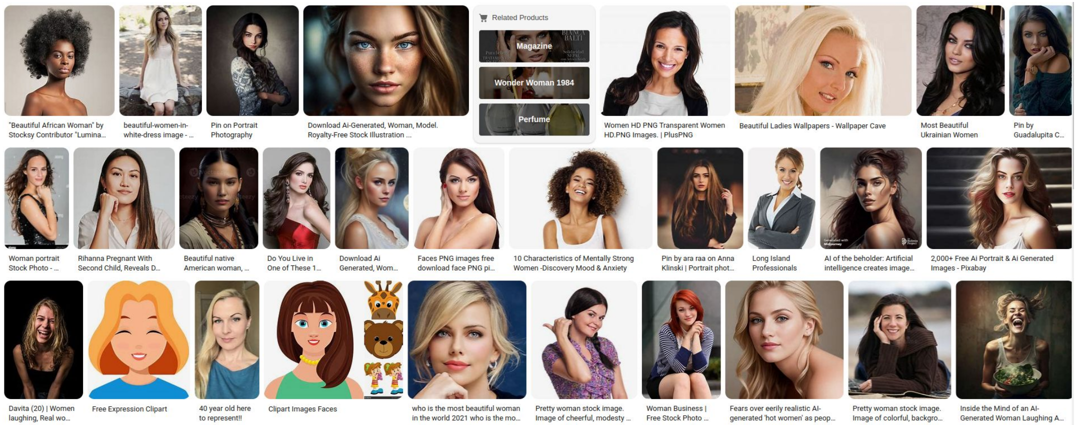
United States

Demographic biases (Bing)

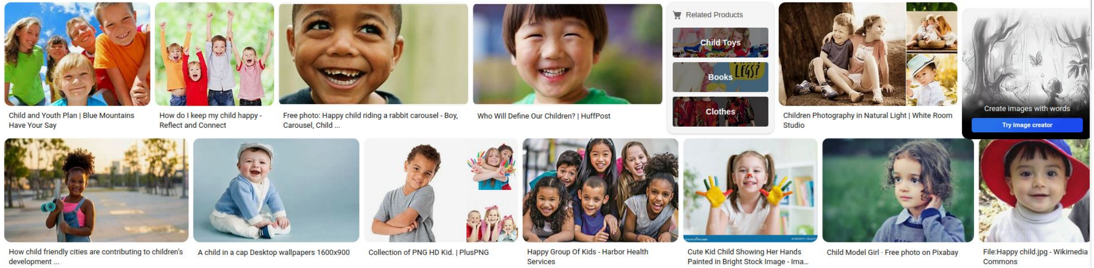
Person



Woman



Child



Framing bias

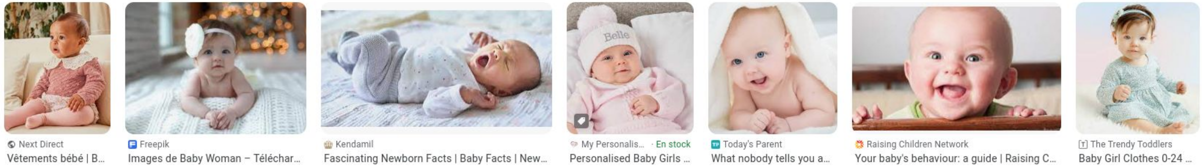
We refer to framing bias as any associations or disparities that can be used to convey different messages and/or that can be traced back to the way in which the visual content has been composed.

- Examples of biased portrayals
  - Minorities, obese persons, babies
- Occurs when taking photos
  - Amplified by labeling and retrieval algorithms by favoring “popular” sources
  - Very detrimental when combined with other biases

Figure 2 Open in figure viewer | PowerPoint

Examples of counter-stereotypical images used in the implicit retraining task.

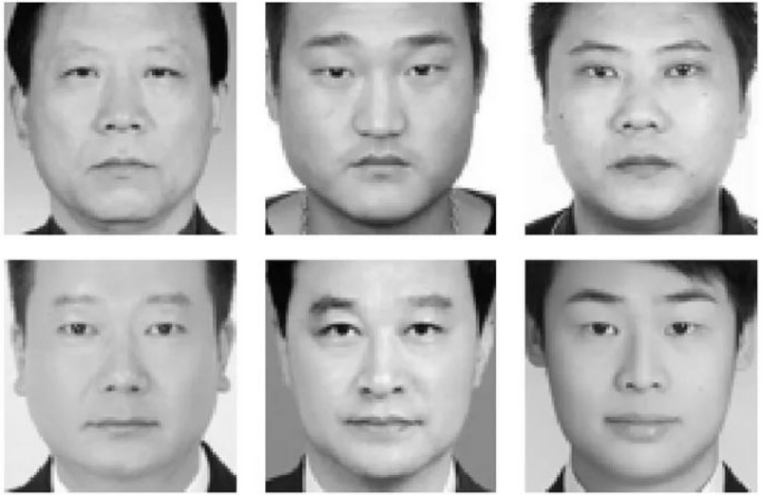
Counter-stereotypical images



Label bias

> *We define* label bias *as any errors in the labelling of visual data, with respect to some ground truth, or the use of poorly defined or inappropriate semantic categories.*

- Causes
  - Intrinsic difficulty of the labeling task
  - Poor semantic definition of the task
  - Limited expertise of participants or annotation models
  - Interpersonal differences between annotators - subjective tasks
- Effects
  - Person-oriented computer vision applications

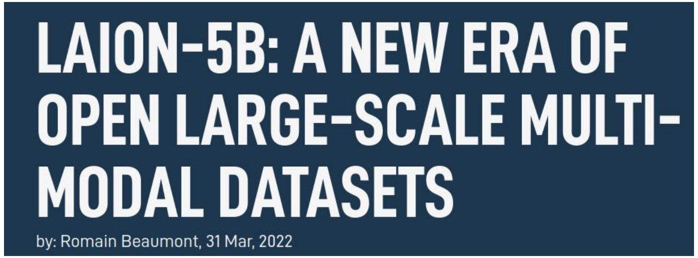


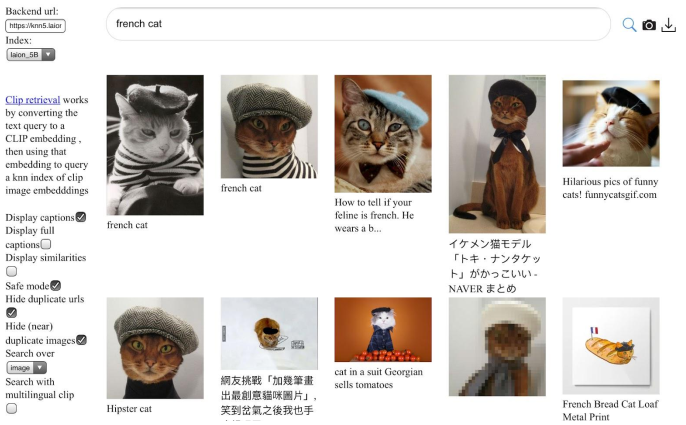
Data-related biases

Dataset examples

LAION dataset

- ImageNet’s younger and significantly larger “brother” project
- Dataset links to images and associated texts from the Web
- Useful to build large multimodal models and compare them
- Not safe for work classifier implemented






Laion-5b: An open large-scale dataset for training next generation image-tex models
C Schuhmann, R Beaumont, R Vencu... - Advances in neural ..., 2022 - proceedings.neurips.cc
... our data collection process for **LAION-5B** in Section 3. Section 4 then describes **LAION-5B**'s ...
Before concluding, we discuss the technical limitations of **LAION-5B** in Section 6 and safety ...
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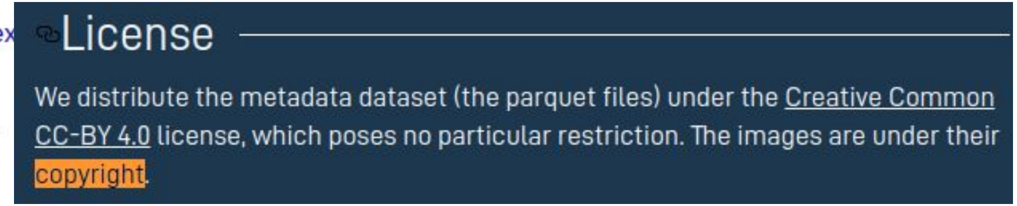

License

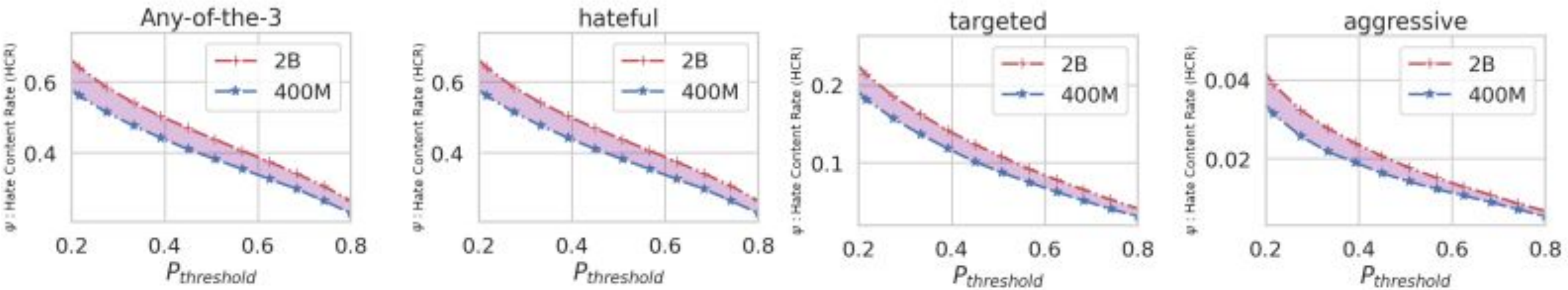
We distribute the metadata dataset (the parquet files) under the Creative Common CC-BY 4.0 license, which poses no particular restriction. The images are under their copyright.

LAION biases

Do the datasets contain images that may be disturbing for viewers?

No, but links in the datasets can lead to images that are disturbing or discomforting depending on the filter or search method employed.

- Metrics to measure language that is hateful, targeted, or aggressive
- Measures based on a pre-existing tool: *pysentimiento*
- Scale effect: more problematic content found in the larger 2B version
- NSFW classifier works for over 99% of the cases, but some hateful, targeted or aggressive content remain



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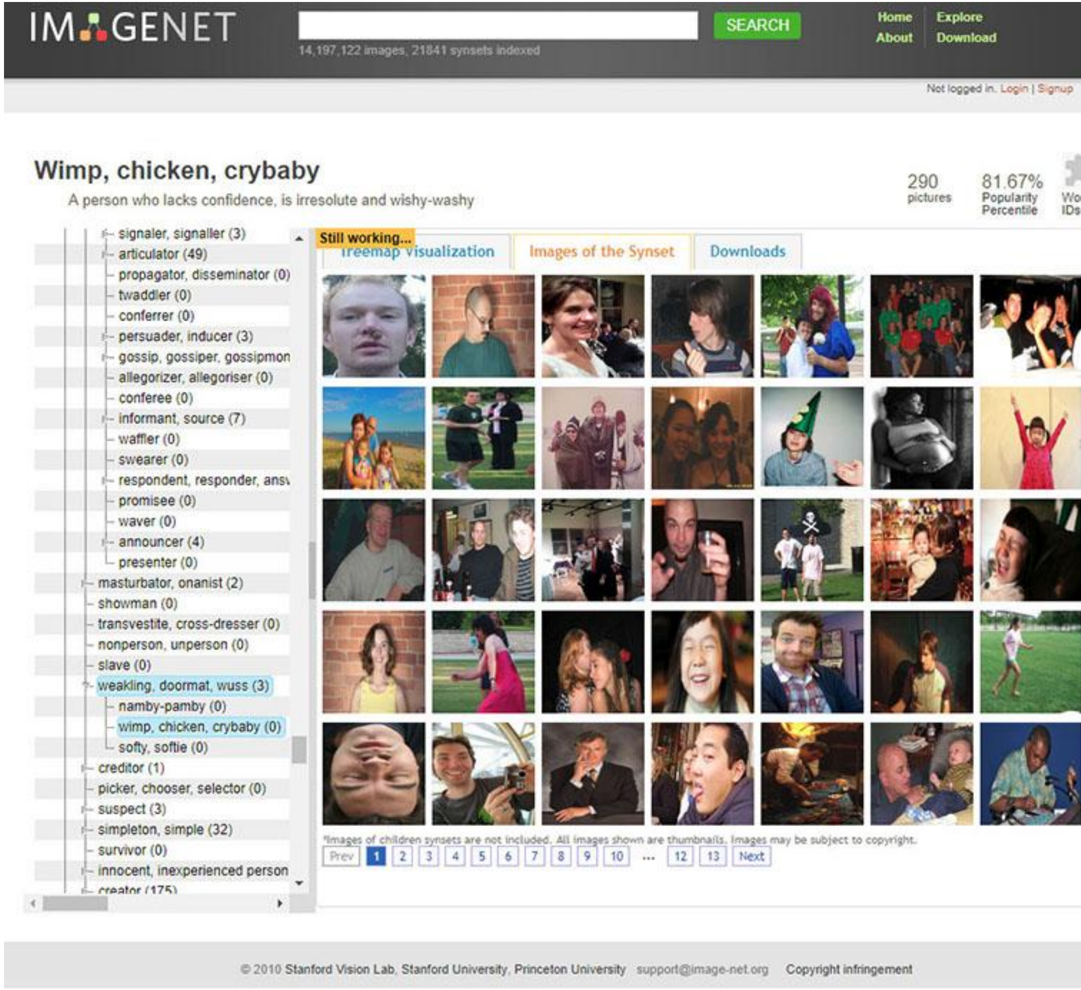
- Focus on person categories from ImageNet
- Impossible to depict most of these classes in a visually coherent and unbiased manner
- Combination of biases
- Image recontextualization
- Reidentification problem
    - +Clearview



Imageability and demographics of ImageNet person classes

Figure 3: (Left) The computer vision model's classification accuracy vs. synset imageability for 143 safe synsets which contain at least 1000 images. More imageable synsets are not necessarily easier for models to recognize, with Pearson correlation coefficient $r = 0.23$. (Right) Example images from synsets that are non-imageable and hard to classify (`conversational partner`); non-imageable but easy to classify (`ancient`); imageable but hard to classify (`groom`); imageable and easy to classify (`black belt`).

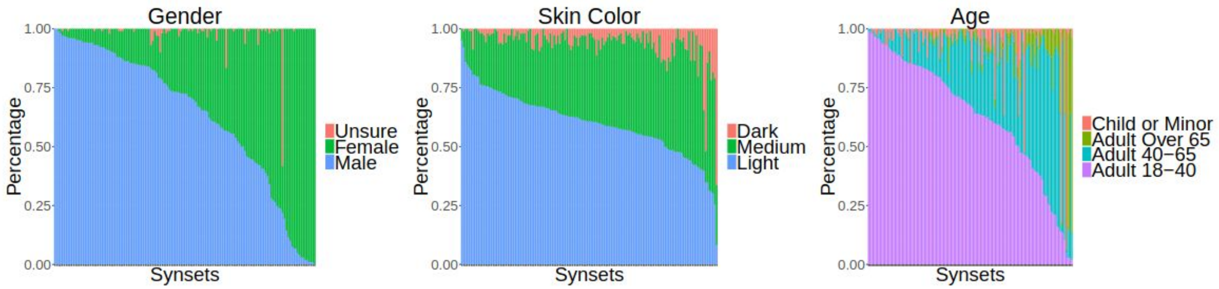
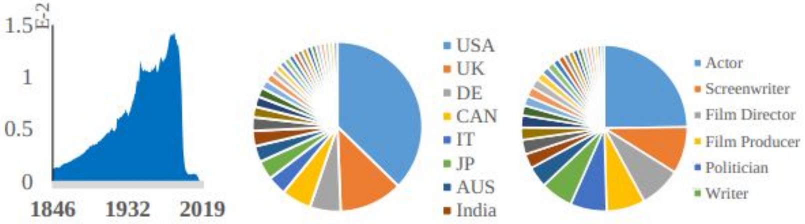


Figure 4: The distribution of demographic categories across the 139 safe and imageable synsets which contain at least 100 images. The size of the different color areas reveal the underrepresentation of certain groups.

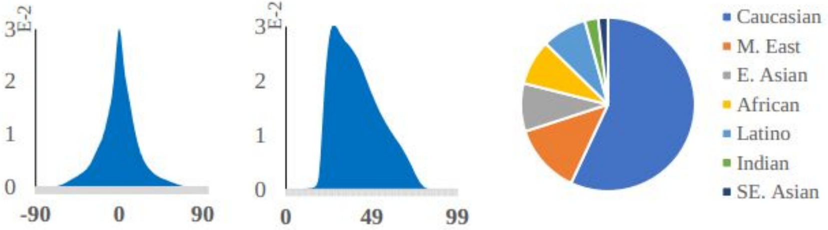
Demographics of WebFace260M

- One of the largest publicly available face datasets
- Curated for supervised learning - using celebrities names from IMDB
- Affected by demographic biases (age, gender, ethnicity) and face presentation biases (pose, size)
- Instances of sampling biases



(a) Date of Birth (b) Nationality (c) Profession

Figure 2: Date of birth, nationality and profession of WebFace260M.

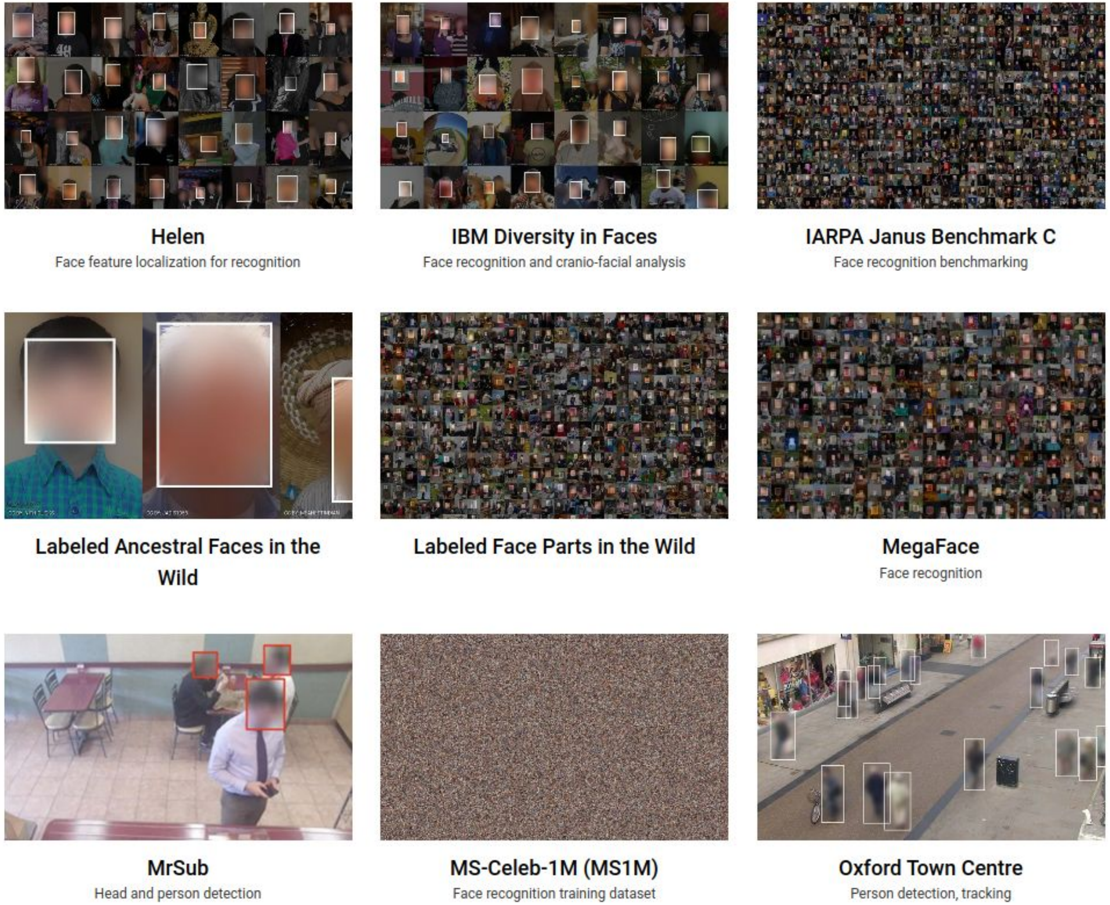


(a) Pose (b) Age (c) Race

Figure 3: Pose (yaw), age and race of WebFace42M.

https://exposing.ai

- “an art and research publication investigating the ethics, origins, and individual privacy implications of face recognition datasets created "in the wild."
- Face search in existing public datasets
- Led to dataset removal



Data-related biases

Text

Biases in textual datasets

- Biases types are similar with those for visual datasets for supervised natural language processing tasks
  - Text classification, sentiment analysis, etc.
- Notable differences for large language models
  - Self-supervised learning with pretext task
  - Text labels are not needed
- Very large datasets that are often unavailable
  - "95% of Mistral's technical expertise is the composition of its database."

    Arthur Mensch (CEO, Mistral AI)

The Pile

- 800G of diversified text
  - Mostly English
- Overrepresentation of
  - Scientific papers
    - Biomedical
  - Code
  - Technical discussions
- Underepresentation of
  - General knowledge
  - Social chats
  - Emails
  -

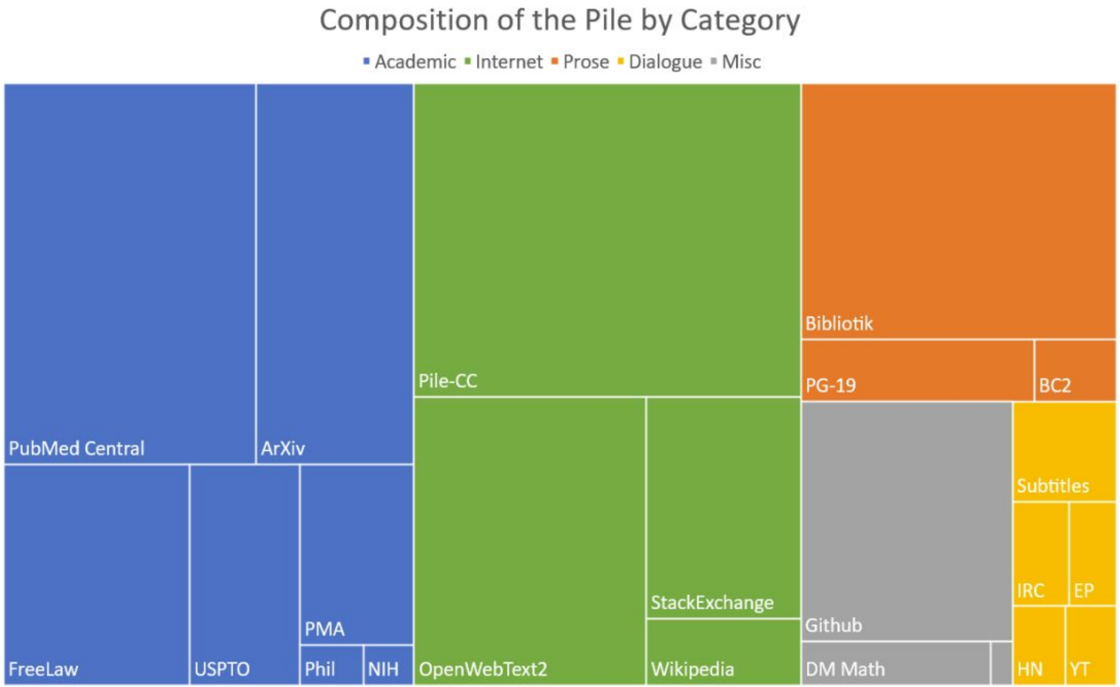


Figure 1: Treemap of Pile components by effective size.

The Pile - Documentation

- Datasheet for the Pile
  - https://arxiv.org/pdf/2201.07311.pdf
- Motivation: training LLMs + motivation per specific dataset
- Dataset composition: description of instances
  - Pile-CC: Instances are webpages; Github: Instances are code files
- Collection process: Web scraping, repository scraping, data dumps
- Data preprocessing: link to an existing paper + raw data accessible
- Dataset distribution: https://github.com/EleutherAI
  - MIT License
  - Copyrighted material distributed under American "fair use" doctrine
- Dataset maintenance: done by EleutherAI
- Legal and Ethical Considerations
  - No ethical review
  - Offensive content: "probably yes"
  - Identification of subpopulations: yes
  - Identification of individuals: not applicable???

The Pile diversity analysis

- Perplexity: captures the degree of ‘uncertainty’ a model has in predicting text
- 16-topic LDA models
- Pile CC as baseline
- New data in PILE for
  - Programming
  - Logic
  - Physics
  - Legal knowledge

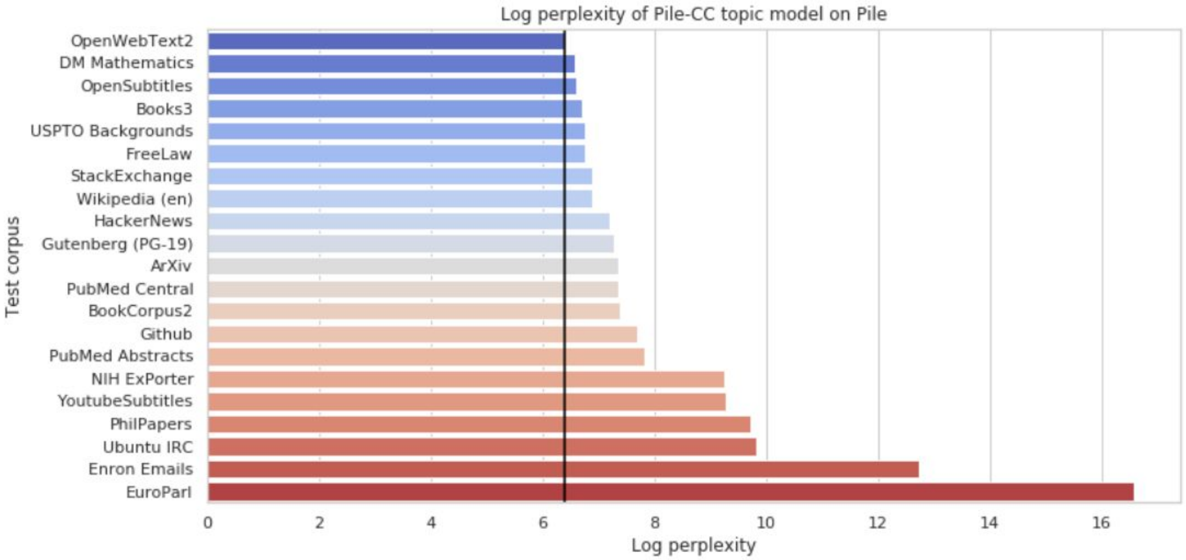
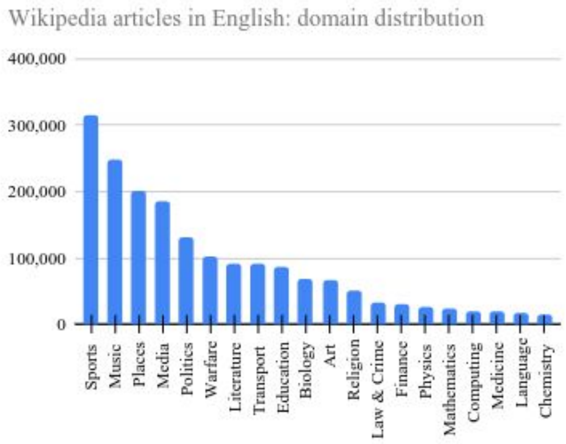


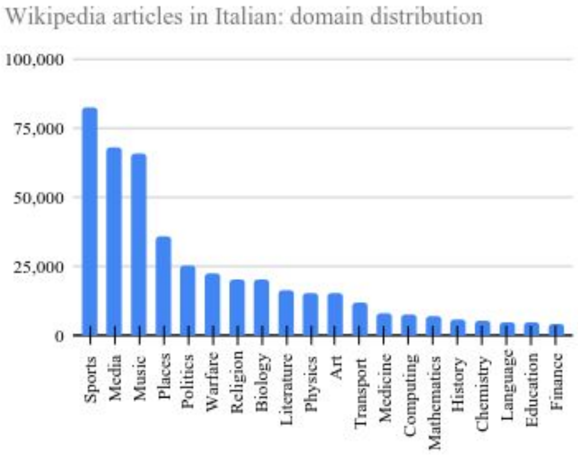
Figure 7: Log perplexity of 16-topic LDA trained on Pile-CC, on other Pile components. Dotted line indicates log perplexity of the topic model on OpenWebText2. Higher indicates a larger topical divergence from Pile-CC.

Biases in LLMs training data

- Selection bias resulting from a given choice of the texts included in training datasets
  - Imbalanced language and culture representation
    - Bias toward high-resource languages
  - Unbalanced distribution of domain and genre
  - Different distributions in different languages
  - Temporal shift in languages
    - Particularly in the long run
  - Demographics of the dataset creators



(a) Domain distribution in the English Wikipedia.



(b) Domain distribution in the Italian Wikipedia.

Bias in a tabular data

- B: 1000 $(Bk-0.63)^2$
  - Bk - percentage of black residents
  - Root problem: racial bias
  - No clear justification of the formula
- LSTAT: lower status
  - Root problem: socioeconomic stereotyping
  - No clear definition of the feature
- Temporal obsolescence
  - Dataset collected in the 1970s

Boston housing dataset

Data Card | Code (154) | Discussion (1)

About Dataset

Domain: Real Estate

Difficulty: Easy to Medium

Challenges:

1. Missing value treatment
2. Outlier treatment
3. Understanding which variables drive the price of homes in Boston

Summary:
The Boston housing dataset contains 506 observations and 14 variables. The dataset contains missing values.

Task-oriented bias analysis

Self-supervised image representation learning

- No labels needed
- DinoV2 - dataset creation
  - Mix of existing datasets built for different visual tasks
  - Similar image retrieval to enrich the initial datasets
- Biases
  - Dominance of classification tasks
  - Large bias toward ImageNet
    - Particularly for the 1k subset
  - Retrieval mostly for points of interest

Task	Dataset / Split	Images	Retrieval	Retrieved	Final
classification	ImageNet-22k / –	14,197,086	as is	–	14,197,086
classification	ImageNet-22k / –	14,197,086	sample	56,788,344	56,788,344
classification	ImageNet-1k / train	1,281,167	sample	40,997,344	40,997,344
fine-grained classif.	Caltech 101 / train	3,030	cluster	2,630,000	1,000,000
fine-grained classif.	CUB-200-2011 / train	5,994	cluster	1,300,000	1,000,000
fine-grained classif.	DTD / train1	1,880	cluster	1,580,000	1,000,000
fine-grained classif.	FGVC-Aircraft / train	3,334	cluster	1,170,000	1,000,000
fine-grained classif.	Flowers-102 / train	1,020	cluster	1,060,000	1,000,000
fine-grained classif.	Food-101 / train	75,750	cluster	21,670,000	1,000,000
fine-grained classif.	Oxford-IIIT Pet / trainval	3,680	cluster	2,750,000	1,000,000
fine-grained classif.	Stanford Cars / train	8,144	cluster	7,220,000	1,000,000
fine-grained classif.	SUN397 / train1	19,850	cluster	18,950,000	1,000,000
fine-grained classif.	Pascal VOC 2007 / train	2,501	cluster	1,010,000	1,000,000
segmentation	ADE20K / train	20,210	cluster	20,720,000	1,000,000
segmentation	Cityscapes / train	2,975	cluster	1,390,000	1,000,000
segmentation	Pascal VOC 2012 (seg.) / trainaug	1,464	cluster	10,140,000	1,000,000
depth estimation	Mapillary SLS / train	1,434,262	as is	–	1,434,262
depth estimation	KITTI / train (Eigen)	23,158	cluster	3,700,000	1,000,000
depth estimation	NYU Depth V2 / train	24,231	cluster	10,850,000	1,000,000
depth estimation	SUN RGB-D / train	4,829	cluster	4,870,000	1,000,000
retrieval	Google Landmarks v2 / train (clean)	1,580,470	as is	–	1,580,470
retrieval	Google Landmarks v2 / train (clean)	1,580,470	sample	6,321,880	6,321,880
retrieval	AmsterTime / new	1,231	cluster	960,000	960,000
retrieval	AmsterTime / old	1,231	cluster	830,000	830,000
retrieval	Met / train	397,121	cluster	62,860,000	1,000,000
retrieval	Revisiting Oxford / base	4,993	cluster	3,680,000	1,000,000
retrieval	Revisiting Paris / base	6,322	cluster	3,660,000	1,000,000
					142,109,386

Table 15: **Composition of our LVD-142M dataset.** We report the list of datasets and associated splits used to build the dataset, how they were included (as is without retrieval or via sample-based or cluster-based retrieval). For retrievals, we indicate the actual number of retrieved images and the final number included in the dataset.

Sentiment analysis for texts

NewsMTSC: A Dataset for (Multi-)Target-dependent Sentiment Classification in Political News Articles

- Sentiment classification is subjective
  - Formulate the task at entity level to have refined text representations
  - Ask annotators to assess from the author's point of view
  - Keep samples that are annotated coherently by several participants
  - Sample sentences from POLUSA and Bias Flipper datasets
- Biases
  - Domain coverage: focus toward the United States and a specific timeframe
    - 30% des exemples avec Donald Trump et/ou Hillary Clinton
  - Difficulty: coherence-based selection removes difficult samples

Real-world Affective Faces Database

Emotion analysis for images

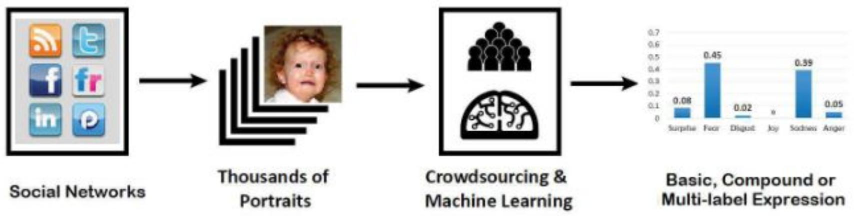
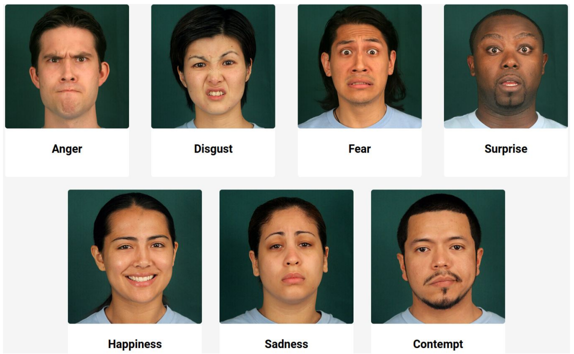


Figure 2. Overview of construction and annotation of RAF-DB.

- Dataset collection
  - Images collected from Flickr
  - Keyword selection with: smile, giggle, cry
  - 315 annotators (students+university staff)
  - ~40 independent labelers per image
- Biases
  - Emotion universality
    - Emotion expression
    - Perception by annotators
  - Data sources are not clear between text and visual overview
  - Gender: 52 / 43 / 5 % distribution for female / male / unsure
  - “Race”: 77% Caucasian, 8% African-American, 15 Asian



Recommender systems biases

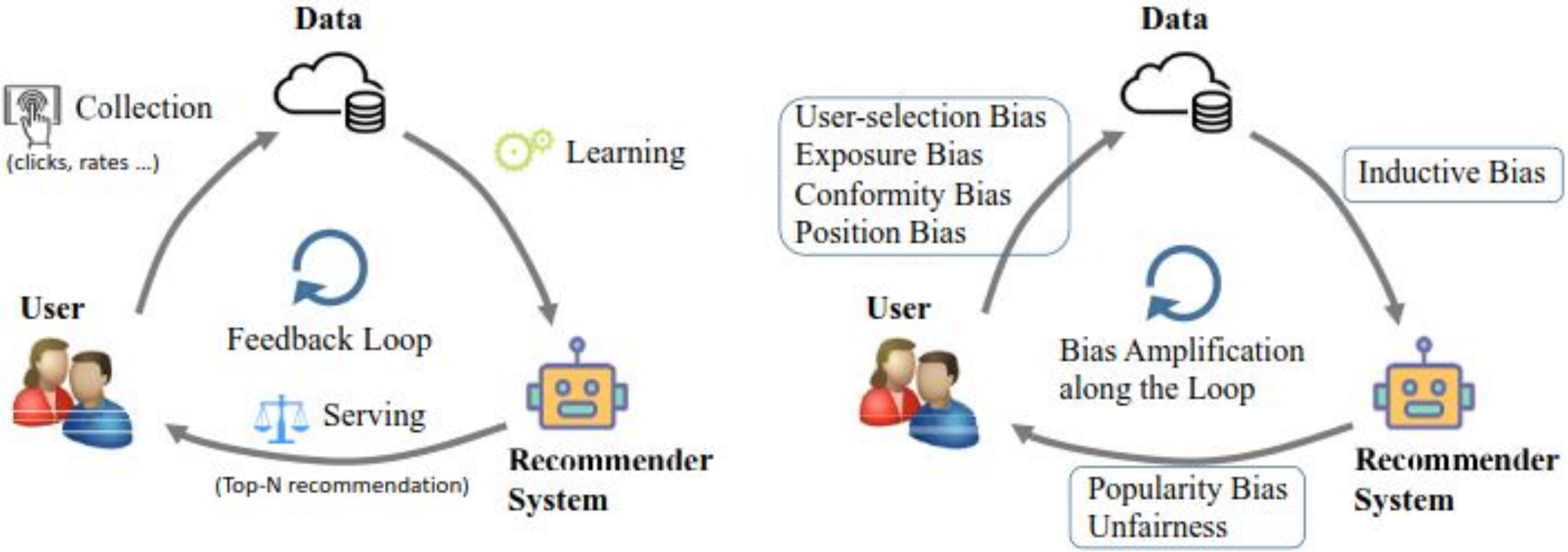


Fig. 2. Feedback loop in recommendation, where biases occur in different stages.

Recommender systems biases

- **Selection**: rated items (observed) are not representative of all ratings
- **Exposure**: unobserved interactions do not always have negative pref.
- **Conformity**: users tend to follow the group, even against own pref.
- **Position**: highly ranked items receive more interactions, regardless of their actual relevance
- **Inductive**: assumption made by the model to better learn the target function and generalize (ex. hard negative oversampling)
- **Popularity**: popular items are recommended too often
- **Unfairness**: discrimination against underrepresented groups

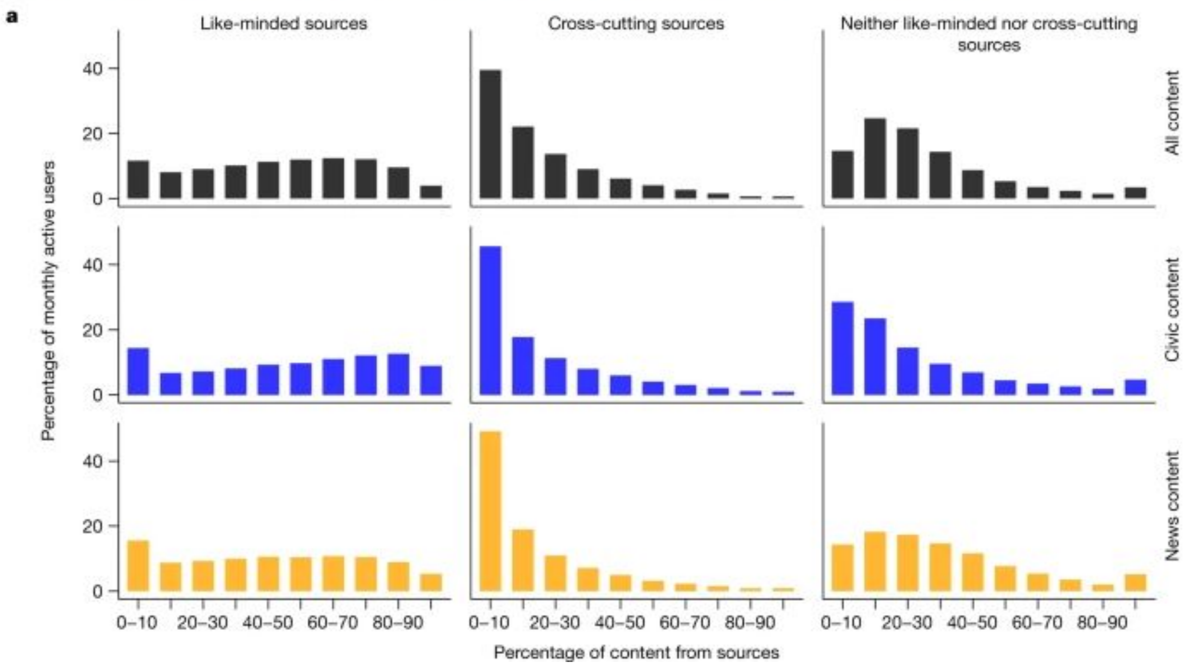
Recommender systems biases

Table 1. The characteristics of seven types of biases in recommendation and the bias amplification in loop.

Types	Stages in Loop	Cause	Effect	Major solutions
Selection Bias	User→Data	Users' self-selection	Skewed observed rating distribution	Data Imputation; Propensity Score; Joint Generative Model; Doubly Robust Model
Exposure Bias	User→Data	Item Popularity; Intervened by systems; User behavior and background	Unobserved interactions do not mean negative	Giving confidence weights by heuristic, sampling or exposure-based model; Propensity Score; Causality-based Model
Conformity Bias	User→Data	Conformity	Skewed interaction labels	Modeling social or popularity effect
Position Bias	User→Data	Trust top of lists; Exposed to top of lists	Unreliable positive data	Click models; Propensity Score; Trust-aware Model
Inductive Bias	Data→Model	Added by researchers or engineers	Better generalization, lower variance or Faster recommendation	-
Popularity Bias	Model→User	Algorithm and unbalanced data	Matthew effect	Regularization; Adversarial Learning; Causal Graph
Unfairness	Model→User	Algorithm and unbalanced data	Unfairness for some groups	Rebalancing; Regularization; Adversarial Learning; Causal Modeling
Bias amplification in Loop	All	Feedback loop	Enhance and spread bias	Break the loop by collecting random data or using reinforcement learning

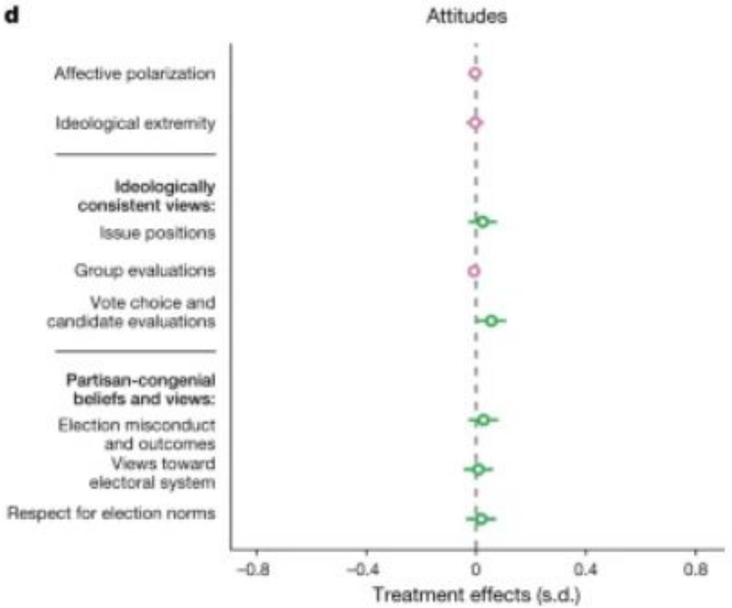
Analysis of echo chambers (filter bubbles) in Facebook

Fig. 1: The distribution of exposure to content among Facebook users.



a, The distribution of the exposure of monthly active adult Facebook users in the USA to content from like-minded sources, cross-cutting sources, and those that fall into neither category in their Facebook

Intervention to reduce filter bubbles



- >23k users who provided consent
  - “Treatment” and control groups
- Downrank content from like-minded users
- Study of behavioral data and surveys
  - 5 surveys before and after the 2020 US elections
- No notable effect of the “treatment” due to reduced exposure to content from like-minded users
  - Regardless of preregistered ideology, political sophistication, digital literacy
  - No effect of age or number of years since joining Facebook
- *Disclaimer: study involving Meta researchers*

Evaluation biases

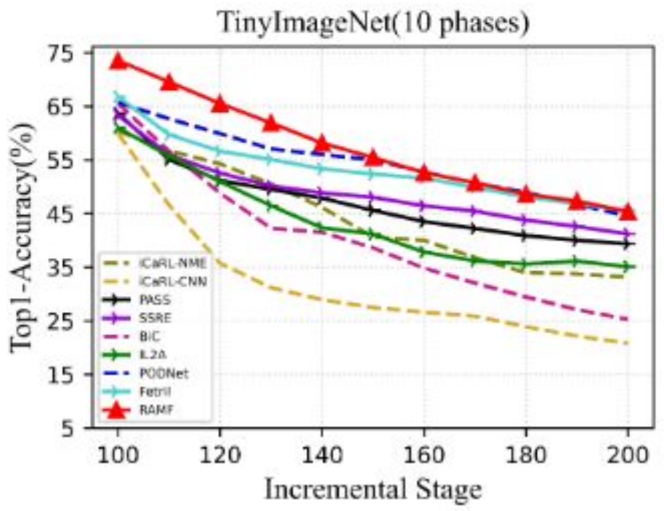
What is fair? - Example of face verification

Model	Sim.	Region = Africa					Region = America					Region = Asia					Region = Europe				
		S. Africa	Nigeria	Egypt	Ghana	Tunisia	US	Canada	Brazil	Argentina	Mexico	India	Japan	S. Korea	Philippines	China	UK	France	Germany	Italy	Ireland
insightface	1	95.65	95.27	93.35	93.23	92.40	95.97	96.21	95.78	97.01	95.51	96.26	93.04	93.75	95.67	95.13	97.17	97.51	97.03	96.80	97.92
	random	97.90	98.56	96.84	98.79	94.11	98.81	98.55	98.84	99.25	100.0	98.84	98.69	99.00	98.63	99.17	99.40	99.66	99.31	99.96	99.48

Face verification accuracy for subjects of different origins

- Using a pretrained model with identities from Wikipedia
- Bias against African and Asian countries
    - More pronounced in some countries than in others
- Optimization of the accuracy metric
    - Similar accuracies per country/region
    - Overall best accuracy by considering the country/region population

Data comparability



- Continual learning: learning with sequential data that do not overlap
  - Add new classes in each learning episode
  - Compute accuracy for all known classes (past and new)

- Strong data augmentation for RAMF (red curve)
  - Pseudo-clases created by mixing actual classes
- Bias:
  - ~10 points initial advantage for RAMF due to data augmentation
  - None of the other methods are tested with the augmented model
  - Compare the slope of the curve rather than the absolute value

Model comparability

- Trained a vision transformer with ~22B params
  - Technically challenging task
  - “We pick a bigger hammer because we can”
- Using a dataset of ~4B images (JFT)
  - Proprietary dataset @ Google
  - Same dataset for all transformer variants
- The evaluation is correct
  - Gains become marginal when changing the parametric footprint 1.8B -> 4.9B -> 22B
- It would be unfair in absence of a discussion of parametric footprint was provided

Table 2: Linear evaluation on ImageNet-1k (Deng et al., 2009) with varying scale. All models pre-trained on large datasets. Performances of a few high-resolution fine-tuned models from are provided for reference.

Model	IN	ReaL	INv2	ObjectNet	IN-R	IN-A
224px linear probe (frozen)						
B/32	80.18	86.00	69.56	46.03	75.03	31.2
B/16	84.20	88.79	75.07	56.01	82.50	52.67
ALIGN (360px)	85.5	-	-	-	-	-
L/16	86.66	90.05	78.57	63.84	89.92	67.96
g/14	88.51	90.50	81.10	68.84	92.33	77.51
G/14	88.98	90.60	81.32	69.55	91.74	78.79
e/14	89.26	90.74	82.51	71.54	**94.33**	81.56
22B	**89.51**	**90.94**	**83.15**	**74.30**	94.27	**83.80**
High-res fine-tuning						
L/16	88.5	90.4	80.4	-	-	-
FixNoisy-L2	88.5	90.9	80.8	-	-	-
ALIGN-L2	88.64	-	-	-	-	-
MaxViT-XL	89.53	-	-	-	-	-
G/14	90.45	90.81	83.33	70.53	-	-
e/14	90.9	91.1	84.3	72.0	-	-

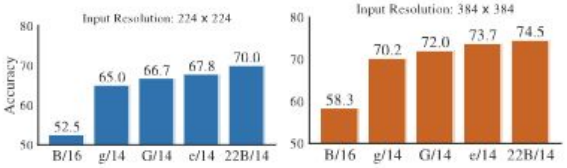


Table 1: ViT-22B model architecture details.

Name	Width	Depth	MLP	Heads	Params(M)
ViT-G	1664	48	8192	16	1843
ViT-e	1792	56	15360	16	3926
ViT-22B	6144	48	24576	48	22165

Evaluation good practices

- State your hypotheses clearly
- Compare the comparable
  - Similar datasets and model sizes (or analyze the impact of differences)
- Use multiple evaluation datasets/tasks
  - Having a sufficient size
  - Focusing on different domains
- Report statistical significance scores
  - Particularly when gains are small
- Avoid train/test overlap
  - Difficult for very large models trained with proprietary datasets

Thank you!
Questions?